Rhaister

Back to basics: Observed statistics are sufficient to predict drug responses

Can summary statistics predict drug responses
in a new cellular context?

Valentine Svensson et al. · Tahoe Therapeutics
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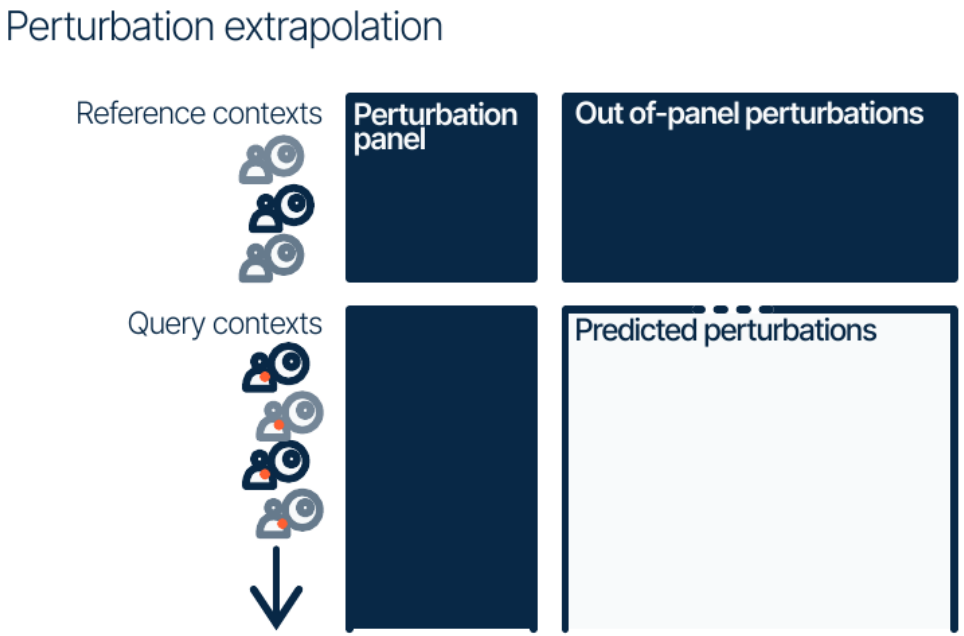
Motivation: the hardest contexts can’t be screened

- We want to predict the **drug** perturbated **gene expression** on rare or unseen **cell lines**
- But we cannot test the drug response on that cell line **directly**:

a) rare cell types
b) Tissues that won’t grow ex vivo
c) A lab is required
d) MONEY and time
Virtual-cell models simulate single cells, then aggregate — slow, costly, no zero-shot, sometimes no better than simple baselines
→ Rhaister inverts this: predict the analysis-ready statistics directly, from the baseline gene expression

Svensson et al., 2026, Introduction, pp. 2–3 and Discussion, p. 9

The task: complete a drug screen in a new context



Known in reference cell lines

Responses to the full drug library

Measured in a new cell line

A panel of drug responses

Predicted

How new cell line responses to the drugs

Tahoe benchmark: 45 reference + 5 query cell lines; 120 panel + 240 target drugs

Svensson et al., 2026, Fig. 1 (top left), pp. 3–4 and 13

What do we have? The data and the notation

Every quantity below is a per-gene statistic over G genes — single cells are never modelled downstream

INPUT · context state

Baseline expression of cell line c

$$x_c = (x_{c,1}, \ldots, x_{c,G}) \in \mathbb{R}^G$$

- Pseudobulk over the untreated (DMSO) cells of line c
- Available for all 50 lines, treated or not
- The only query-line measurement Rhaister-O needs

TARGET · perturbation response

Response of line c to perturbation p

$$y_{c,p} = (y_{c,p,1}, \ldots, y_{c,p,G}) \in \mathbb{R}^G$$

- Treated vs. plate-matched DMSO, computed per line and plate
- p = (drug, dose): every dose is a separate perturbation
- Three views per gene: expression delta Δ, $\log_2$ fold change, MWU p-value

Tahoe-100M: 50 cancer cell lines × 384 drugs × 3 doses (0.05, 0.5, 5 µM), 24 h readout
Split: 45 reference / 5 query lines · 120 panel + 240 target drugs → 360 + 720 perturbations

Svensson et al., 2026, Data, pp. 13–14; summary statistics Eqs. 19–20, p. 14

Rhaister learns a weighted sum of panel responses

Fit the weights on the 45 reference lines, then apply them in the query line

1 · FIT — one ridge problem per target drug p★, solved across reference lines only

$$\widehat{W} = \arg\min_{W \in \mathbb{R}^{|D_x| \times G}} \sum_{c \in \mathcal{C}_{\text{ref}}} \| Y_c - W^\top X_c \|_2^2 + \lambda \|W\|_F^2$$

X = panel responses of the 45 reference lines (45 × |Dₓ| × G) · Y = their responses to the target drug p★ (45 × G) · λ = ridge penalty

2 · PREDICT — a weighted sum of the panel responses measured in the query line c★

$$\hat{y}^{\text{ridge}}_{c^\star, p^\star, g} = \sum_{q \in D_x} \widehat{W}_{q,g} \; y_{c^\star, q, g}$$

ŷ = the predicted statistic: one gene, one target drug, one query line.
y(c★, q, g) = the panel response already measured in the query line — the only query-line data Rhaister uses.

Fitted separately for each view: expression delta, $\log_2$ fold change, and MWU significance

Rhaister leads the Tahoe-100M comparison

Held-out cell lines, measured drug panel, five splits

Model	Discrim.	Pearson Δ	PR-AUC	LFC rank	DE overlap	ES rank
Rhaister	**0.96**	**0.87**	**0.73**	**0.81**	**0.59**	**0.96**
STATE	0.93	0.81	0.51	0.57	0.40	0.89
Additive	0.82	0.55	0.55	0.61	0.43	0.85
Half-sample	0.93	0.77	0.62	0.67	0.52	0.97

Higher is better. Values are reported means; SD bars omitted from this table.

The half-sample reference is an empirical comparison, not a hard ceiling.

Svensson et al., 2026, Fig. 1, Tahoe-100M panel; p. 12, half-sample reference

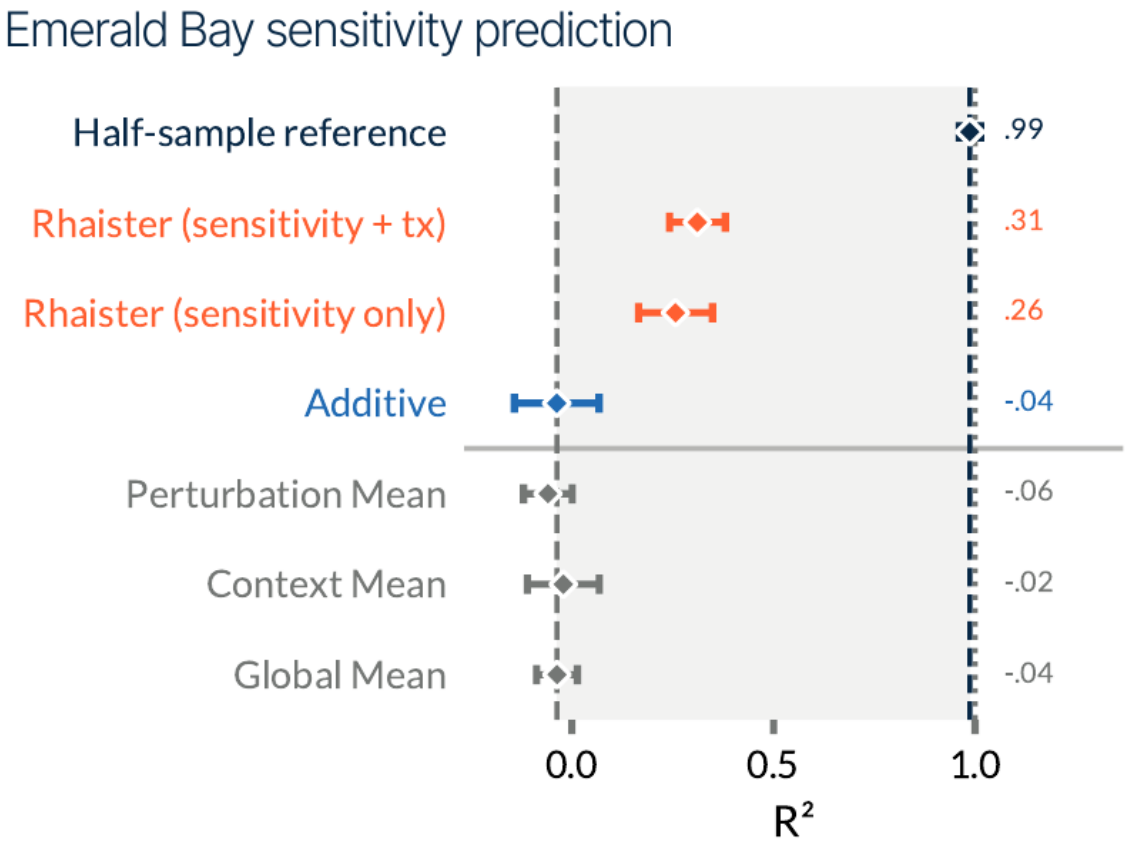
Transcriptomic responses improve sensitivity prediction

Emerald Bay

52 pooled cancer cell lines
5-day drug treatment
Paired abundance and RNA readouts

0.26 to 0.31

R^2 after adding transcriptomic responses from the measured panel



Endpoint: change in relative cell-line abundance within a pooled culture

Rhaister-O predicts from baseline expression alone

Zero-shot: no treated cells from the query line

1 · FIT — per target drug p, per gene g, over the 45 reference lines

$$\left(\hat{\beta}_{p,g},\ \hat{\gamma}_{p,g}\right) = \arg\min_{\beta,\,\gamma} \sum_{c \in \mathcal{C}_{\mathrm{ref}}} \left(y_{c,p,g} - \beta - \gamma\, x_{c,g}\right)^2$$

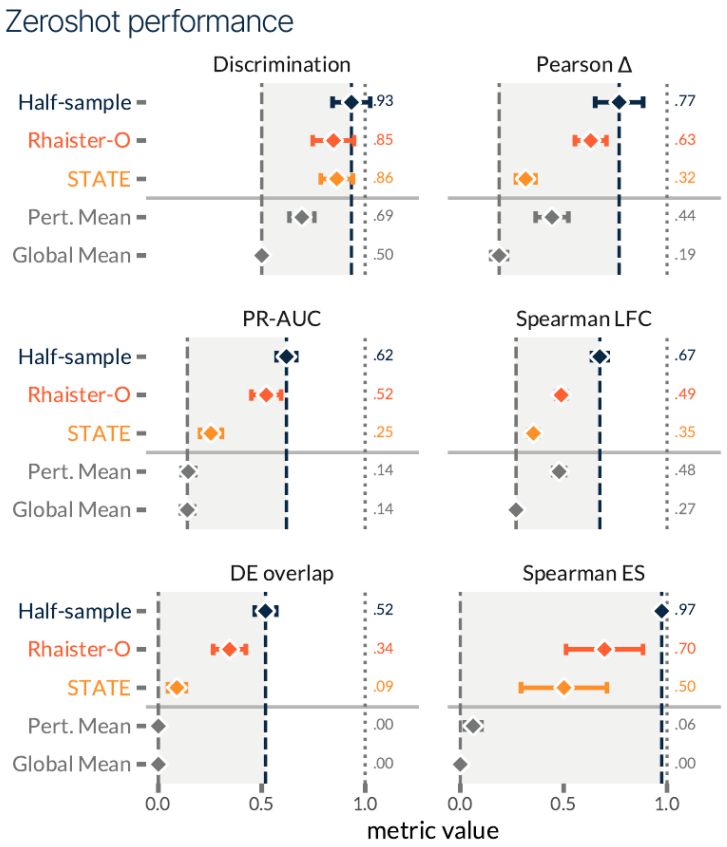
2 · PREDICT — query line c★, unseen target drug p★

$$\hat{y}_{c^\star,p^\star,g} = \hat{\beta}_{p^\star,g} + \hat{\gamma}_{p^\star,g}\ x_{c^\star,g}$$

Two parameters per (drug, gene), fitted on 45 points — one per reference line.
x(c★, g) is the only query-line input, and it is untreated.

Pearson Δ: 0.63 vs 0.32

Rhaister-O compared with STATE
Discrimination: 0.85 vs 0.86



Summary statistics provide a strong baseline

The contribution

Direct prediction of the readouts that downstream analyses use

The boundary

Known perturbations, new contexts
Population summaries leave single-cell heterogeneity unresolved

For therapeutic discovery: useful prioritization still needs experimental validation

Svensson et al., 2026, Discussion, p. 9; interpretation of Figs. 1–4

Cross-dataset gains depend on the metric

Selected metrics from the additional perturbation screens

Screen / metric	Rhaister	STATE	Additive
Parse PBMC: Pearson Δ	0.46	0.25	0.43
Parse PBMC: LFC rank	0.59	0.34	0.63
Replogle–Nadig: Pearson Δ	0.41	0.37	0.39
Replogle–Nadig: LFC rank	0.63	0.78	0.66

PRISM sensitivity: Rhaister R^2 0.87; perturbation mean R^2 0.81

A high absolute score can coexist with a strong context-independent baseline.

ID: 200010263

Panel size and reference diversity affect performance

Measured perturbation panel

1–120 drugs

Three doses per drug
Larger panels improve the six metrics

Reference contexts

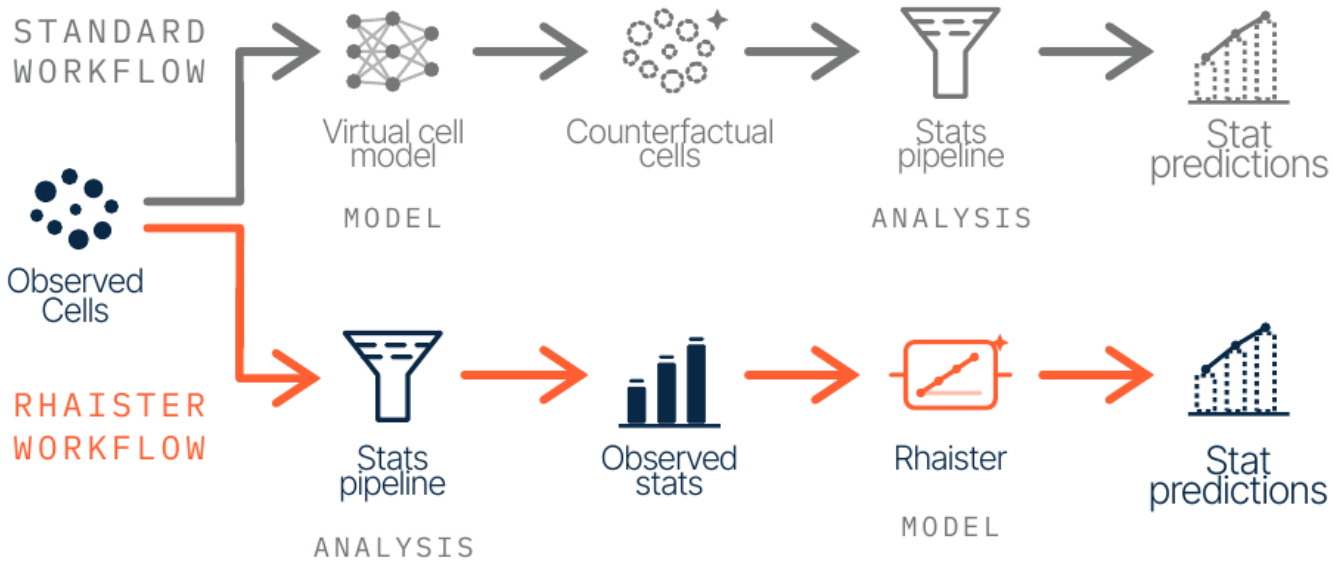
~20 cell lines

Marginal further gains in the panel-based setting

Baseline-only Rhaister-O continues to benefit from more reference contexts.

ID: 200010263

The full framework adds fallback and calibration



Additive fallback

Used when reference coverage is insufficient

MWU calibration

Small MLP, then an equal blend with the ridge prediction

Predicted significance statistics do not quantify uncertainty in the model prediction.

Svensson et al., 2026, Fig. 1 top right; Details, p. 11, Eqs. 1–7

Metric definitions and source

Metric	What it evaluates
Discrimination	Rank of the correct perturbation among alternatives
Pearson Δ	Agreement of expression-change magnitudes across genes
PR-AUC / DE overlap	Differential-expression classification / gene-set agreement
Spearman LFC	Rank agreement of fold changes in significant genes
Spearman ES	Rank of the number of significant genes across perturbations

Source version: supplied PDF, posted 12 June 2026 (preprint)

Model and code: https://huggingface.co/tahoebio/Rhaister
Full definitions: paper pp. 14–15. Source discrepancies are documented in the speaker notes.